

**DATA VISUALIZATION ASSOCIATE
EARLY REMOTE INTERNSHIP**

Week 1: Data Setup & Initial EDA Report

*Database Environment Configuration and Exploratory Analysis of the Opportunity and Learner
Datasets*

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Deliverable: Week 1 Data Setup & Initial EDA Report

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1. Introduction

1.1 Purpose of Week 1

The Week 1 of the Data Visualization Associate Early Remote Internship establishes the technical and analytical foundation for subsequent engagement available in the module. The primary objective was to put up a working PostgreSQL database environment, import the two datasets that will drive all subsequent analysis and dashboard work (Opportunity Data and Learner Data) and perform an initial round of exploratory data analysis (EDA) sufficient to understand the shape, structure, and reliability of each dataset before any cleaning or modelling begins.

1.2 Objectives

- Set up a PostgreSQL database environment
- Use any of the guides as the managed hosting provider (Supabase, Aiven, Spreadsheet).
- Import the Opportunity Data and Learner Data datasets and verify successful, complete uploads.
- Review table structure, field names, and data types for both datasets.
- Establish a working connection between the Aiven PostgreSQL and Looker Studio.
- Use SQL queries to explore dataset size, distributions, categories of opportunities available, and completeness.
- Identify missing values or inconsistencies, and potential data quality issues.
- Document initial observations and insights that will guide Week 2 cleaning and integration work.

1.3 Importance of Database Setup

A well configured database environment is the foundation on which every later deliverable in this internship depends. Errors introduced at the setup stage have already truncated data. A proper database setup ensures data integrity, security, and scalability. It prevents data corruption, eliminates redundancy, and allows multiple users to access information simultaneously. Setting up your database correctly from the start optimizes query performance and protects critical information from system failures or unauthorized access.

1.4 Importance of Exploratory Data Analysis

Exploratory data analysis is the foundation of a reliable data science pipeline that researchers determine if the data is high quality and free of missing values, and ensure the statistical techniques they plan to use are appropriate for the specific dataset. It helps to prevent costly modelling mistakes later on. (IBM)

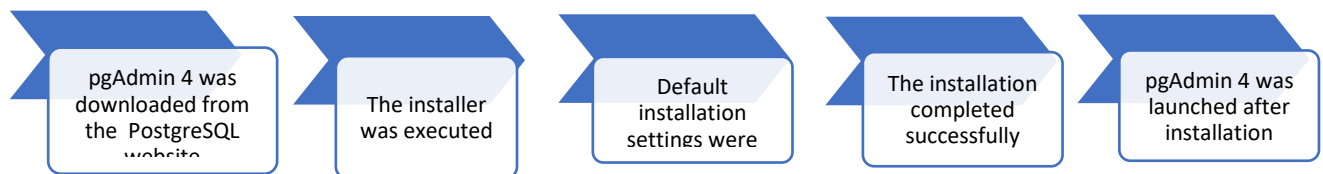
2. Database Setup

This report documents the complete process of setting up a cloud-hosted PostgreSQL database using Aiven, connecting the database to pgAdmin 4, uploading datasets into PostgreSQL, and verifying that the datasets were successfully loaded. The report also outlines the connection method used to establish secure communication between the local PostgreSQL client and the cloud database.

The purpose of this setup was to prepare a centralized PostgreSQL database that can later be connected to analytics and visualization tools such as Google Looker Studio for reporting and dashboard creation.

2.1 PostgreSQL Installation and Database Creation

Installing pgAdmin 4:

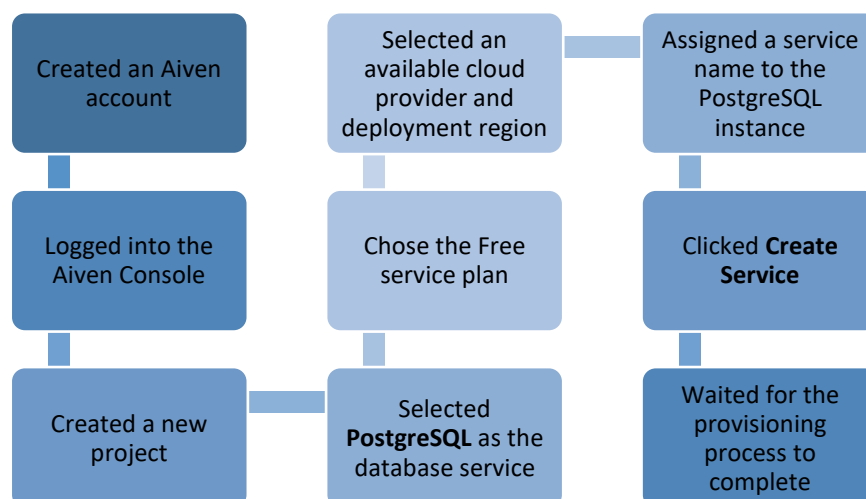


Upon launch, pgAdmin 4 provided the interface for registering and managing PostgreSQL servers.

2.2 Using Aiven PostgreSQL

Since a cloud-hosted PostgreSQL instance was required, Aiven was selected because it offers a fully managed PostgreSQL service that simplifies database operations. It provides automatic backups, high availability, seamless scaling, built-in security, monitoring, multi-cloud deployment, and native compatibility with Looker Studio's PostgreSQL connector, which made it the most direct path to the Week 1 deliverable.

Steps Performed:



After deployment, the PostgreSQL instance became active and ready for use. Once the database service was successfully created, the connection details were retrieved from the Aiven dashboard. The following information was obtained: Service URI, Host Name, Port Number, Database Name, Username, Password and SSL Mode. The connection information was later used to connect pgAdmin 4 and Looker Studio.

Connecting Aiven PostgreSQL to pgAdmin

In a bid to administer the cloud database locally, a server connection was created in pgAdmin. The connection was done following these steps below:

Step 1: opened the pgAdmin4

Step 2: Right-click **Servers**.

Step 3: **Create** → **Server** option was selected. Then named server, Under the **General** tab: a descriptive server name: Excelerate Internship – Aiven. This name serves only as a local identifier inside pgAdmin.

Step 4

Opened the **Connection** tab.

Entered the connection details obtained from Aiven.

Field	Value
Host	Aiven PostgreSQL Host
Port	14688
Maintenance Database	defaultdb
Username	Avnadmin
Password	Aiven-generated password

Saved the configuration.

Database Setup

After successfully connecting to PostgreSQL:

- The default PostgreSQL database (**defaultdb**) became accessible.
- Database objects such as Schemas, Tables, Roles, and Tablespaces were visible.
- The PostgreSQL environment was fully operational and ready for data storage.

This confirmed that the database environment had been configured successfully.

2.3 Dataset Upload Process

The objective of this stage was to upload the provided dataset into PostgreSQL for storage and analysis.

The Opportunity Data source file was imported into a table named `opportunity_data`. The source file's headers include several mixed-case field names for example `Badge`, `CareerAddOn`, `Cohort`, `DropoutTransaction`, `Eligibility`, `NotStartedTransaction` etc. alongside standard lowercase fields such as

category, fee, and duration. Because PostgreSQL folds unquoted identifiers to lowercase by default, these mixed-case columns were imported using double-quoted identifiers to preserve their original casing.

The **Learner Data** source file was imported into a table named learner_opportunity. This file is substantially wider than the Opportunity Data source, containing 43 fields with data plus a large number of empty columns (213) generated by the original spreadsheet export. These blank 213 columns were excluded at import time rather than loaded, since they carried no data and were responsible for upload failure. The upload process involved the following steps.

Step 1: Open the target PostgreSQL database inside pgAdmin.

Step 2: Created the database tables for the two datasets provided (opportunity data & learner opportunity) using SQL scripts.

Step 3: Prepared the dataset in the required format for PostgreSQL import.

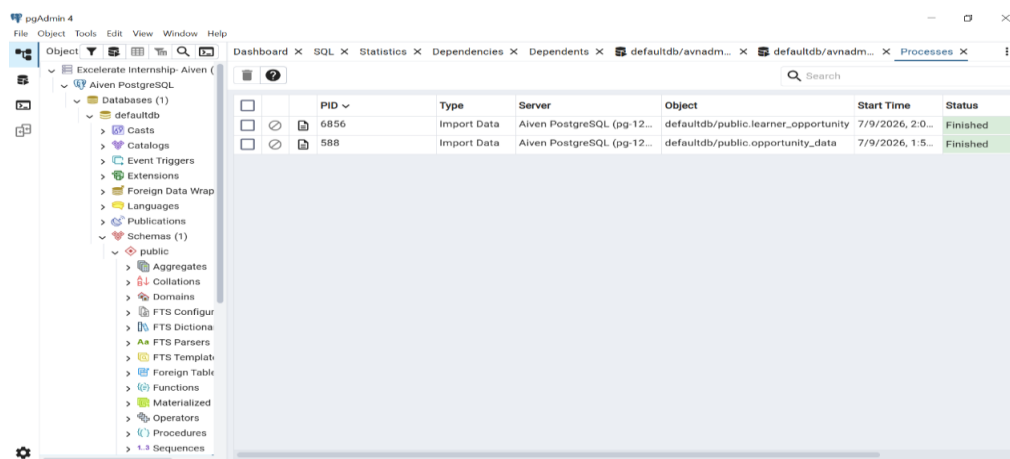
Step 4: Imported the dataset into the corresponding PostgreSQL tables using pgAdmin's Import/Export functionality.

Step 5: Verified that the import process completed without errors.

Confirmation of Successful Dataset Upload

The dataset upload was considered successful after completing the following verification checks:

- Successful connection to PostgreSQL.
- SQL execution completed successfully.
- Database objects remained accessible.
- Uploaded tables were available inside the PostgreSQL database.
- Records were stored without import errors.
- The PostgreSQL database was ready for downstream reporting and visualization.



These checks and processes confirmed that the datasets had been successfully loaded into PostgreSQL and were available for querying.

2.4 Verifying Successful Imports

After loading, row counts were checked against the source files to confirm complete, lossless imports:

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including the 'public' schema. The main query window contains the following SQL query:

```
1 SELECT 'opportunity_data' AS table_name, COUNT(*) AS record_count FROM opportunity_data
2 UNION ALL
3 SELECT 'learner_opportunity', COUNT(*) FROM learner_opportunity;
```

The 'Data Output' tab shows the results of the query:

	table_name	record_count
1	learner_opportun...	15029
2	opportunity_data	5733

The status bar at the bottom indicates 'Total rows: 2' and 'Query complete 00:00:06.969'.

Row-count verification queries run immediately after import.

Table	Source File Rows	Rows Loaded	Status
opportunity_data	5,733	5,733	Match import
learner_data	15,029	15,029	Match import

Table 1. Import verification comparing source row counts to loaded row counts.

The screenshot shows the pgAdmin 4 interface with a query window displaying sample records from the 'learner_opportunity' table. The query is:

```
1 SELECT *
2 FROM learner_opportunity
3 LIMIT 5;
```

The 'Data Output' tab shows the first five rows of the data:

	pk	sk	assigned_cohort	fee	status	accept_reject_date	amount_to_be_paie	transaction_id	payment_status	Lagree
1	Learner#...	Learner#000000001...	BN0UUUG	[null]	1070	1694156764463	0	[null]	unpaid	[null]
2	Learner#...	Learner#000000001...	[null]	[null]	1030	1696504116486	0	[null]	unpaid	[null]
3	Learner#...	Learner#000000001...	BM48Si8	[null]	1120	1714894790552	0	[null]	unpaid	[null]
4	Learner#...	Learner#000000001...	BOQ05VE	[null]	1055	1693377074111	0	[null]	unpaid	[null]
5	Learner#...	Learner#000000001...	[null]	[null]	1010	[null]	1	[null]	unpaid	[null]

The status bar at the bottom indicates 'Total rows: 5' and 'Query complete 00:00:08.493'.

learner_opportunity Sample records

pgAdmin 4

FileObjectToolsEditViewWindowHelp

Object

Excelerate Internship- Aiven (1)

Aiven PostgreSQL

Servers

Query

Query History

SELECT *

FROM opportunity_data

LIMIT 5;

Data Output

Messages

Notifications

Showing rows: 1 to 5

Page No: 1

of 1

pk	opportunity_id	Badge	CareerAddOn	category	code	Cohort	created_at	currency_type	current_editor	Dr te
1	Opportu...	Opportunity#00...	('sk': 'Badge#000...	NULL	Competiti...	M636023	('sk': 'Cohort#0...	1.66E+12	USD	amrit@vemp...
2	Opportu...	Opportunity#00...	('sk': 'Badge#000...	NULL	Career	A463411	('sk': 'Cohort#0...	1.66E+12	USD	NULL
3	Opportu...	Opportunity#00...	NULL	NULL	Career	A8848RT	('sk': 'Cohort#0...	1.71E+12	NULL	NULL
4	Opportu...	Opportunity#00...	('sk': 'Badge#000...	NULL	Internship	I882052	('sk': 'Cohort#0...	1.67E+12	USD	NULL
5	Opportu...	Opportunity#00...	('sk': 'Badge#000...	NULL	Career	AYVMV...	('sk': 'Cohort#0...	1.72E+12	USD	NULL

Total rows: 5Query complete 00:00:13.961CRLF Ln 3, Col 9

opportunity_data sample records

2.6 Connecting PostgreSQL to Looker Studio (Aiven Method)

Looker Studio connects to Aiven PostgreSQL using its native PostgreSQL connector. The Aiven service's host, port, database name, and credentials were entered directly into Looker Studio's connection dialog, with SSL mode enforced to match Aiven's connection requirements. Once authenticated, both `opportunity_data` and `learner_data` were available as selectable tables and confirmed to return live row counts.

Looker Studio is now called Data Studio. For more, see [our blog](#).

Untitled Report

File Edit View Insert Page Arrange Resource Help

Reset Share View

Data sources

Name	Connector Type	Type	Used in report	Status	Actions	Alias
learner_opportunity	PostgreSQL	Embedded	0 charts, 0 variables	Working	EDIT DUPLICATE REMOVE MAKE REUSABLE	ds1
opportunity_data	PostgreSQL	Embedded	0 charts, 0 variables	Working	EDIT DUPLICATE REMOVE MAKE REUSABLE	ds0

ADD A DATA SOURCE

Figure 1. End-to-end workflow from PostgreSQL provisioning through Looker Studio connectivity.

3. Dataset Overview

3.1 Opportunity Data

The Opportunity Data table (`opportunity_data`) holds information opportunities offered to learners through the Excelerate platform which are, internships, career opportunities, competitions, courses, events, engagements, masterclasses, and job simulations. It contains 5,733 records across 33 fields and functions as the catalogue against which learner applications in `learner_data` are matched.

Purpose

This dataset defines what the Excelerate platform offers: what opportunities exist, what category and duration they fall under, what fee or reward structure applies, and metadata such as eligibility criteria, cohort assignment, and descriptive text used in listings.

Key Fields

Field	Description	Observed Values
<code>opportunity_id</code>	Unique identifier for each opportunity	5,733 unique values
<code>category</code>	Opportunity type	Internship, Career, Competition, Course, Event, Engagement, Masterclass....
<code>fee</code>	Fee associated with the opportunity	Mostly 0; contains extreme outlier values
<code>duration / duration_type</code>	Length of the opportunity	Numeric duration paired with a unit (weeks, months, days, hours)
<code>location</code>	Delivery mode / location	Highly inconsistent — see Section 6
<code>currency_type</code>	Currency for fee amounts	USD, INR, EUR (plus inconsistent variants)
<code>microscholarship</code>	Scholarship or stipend value associated with the opportunity	Wide numeric range, dominated by a single value (120)
<code>name</code>	Opportunity title	Free text; contains test/placeholder entries — see Section 6

Field	Description	Observed Values
Reward	Structured reward metadata (JSON-like)	Present for ~93.6% of records

Table 2. Key fields in the Opportunity Data dataset.

The screenshot shows the pgAdmin 4 interface. On the left, the 'defaultdb' database is expanded, showing various objects like Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages, Publications, Schemas, Subscriptions, Login/Group Roles, and Tablespaces. The 'Query' tab is active, displaying the following SQL query:

```

1 SELECT column_name
2 FROM information_schema.columns
3 WHERE table_name = 'opportunity_data'
4 ORDER BY ordinal_position;

```

The 'Data Output' tab shows the results of the query, listing 12 columns from the 'opportunity_data' table:

column_name	name
1	pk
2	opportunity_id
3	Badge
4	CareerAddOn
5	category
6	code
7	Cohort
8	created_at
9	currency_type
10	current_editor
11	DropoutTransaction
12	duration

The status bar at the bottom indicates 'Total rows: 33' and 'Query complete 00:00:07.829'.

Fields from opportunity_data

Important Variables identified

- category
- fee
- currency_type
- duration
- duration_type
- Reward
- location

3.2 Learner Data

The Learner Data table (learner_opportunity) captures individual learner applications and their progress against opportunities. It contains 15,029 records across 43 meaningful fields (after excluding 213 blank trailing columns carried over from the original spreadsheet export).

Purpose

This dataset represents the learners application information on the platform. It gives insights to who applied, to what opportunity, when, application current status, whether a fee or scholarship amount applied, and how they heard about the opportunity.

Key Fields

Field	Description	Observed Values
status	Numeric application status code	14 valid codes, mapped via a status lookup table
category	Opportunity category associated with the application	Mirrors opportunity_data category values
apply_date	Application submission time	ISO 8601 string format; 60.2% populated
completion_date / reward_awarded_date	Completion and reward time	
payment_status	Payment status of the application fee	paid, unpaid, or null
amount_to_be_paid	Fee amount owed by the learner	81.0% of populated values are 0 (scholarship-covered)
from_where_did_you_hear_about_us	Acquisition / referral source	Free text, high cardinality with duplicate spellings
withdraw_date / dropped_out_date / not_started_date	Funnel exit timestamps	Sparse; consistent with a minority of applicants exiting early
team_name / team_code	Team-based application grouping	Populated only for team-format

Table 3. Key fields in the Learner Data dataset.

Important Variables

- status
- category
- apply_date
- completion_date

- payment_status
- amount_to_be_paid
- referral source
- reward_awarded_date

4. SQL Exploration & Visualization

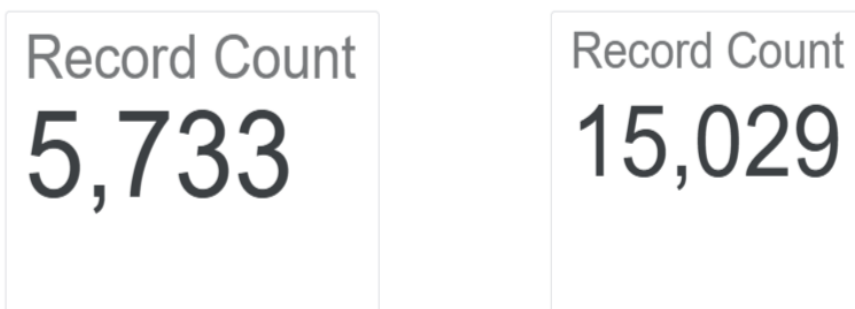
All queries below were run against the live `opportunity_data` and `learner_data` tables on Aiven. Mixed-case column names inherited from the Opportunity Data source (Badge, Cohort, Reward, and similar) are wrapped in double quotes throughout, since PostgreSQL treats unquoted identifiers as case-insensitive and would otherwise fail to resolve them.

4.1 Dataset count

```
SELECT COUNT (*) AS total_records  
FROM opportunity_data;
```

```
SELECT COUNT (*) AS total_records  
FROM learner_opportunity;
```

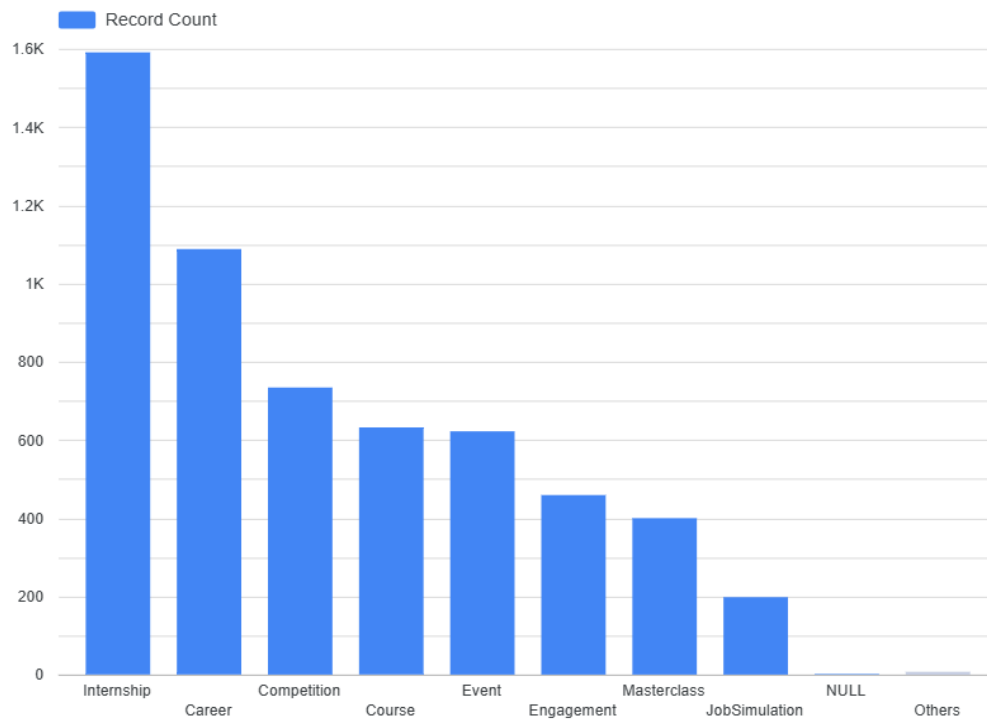
A basic row count confirms both dataset and that the import matches the source files (see Table 1).



Record count Visualization

4.2 Category

```
SELECT category, COUNT (*) AS total
FROM opportunity_data
GROUP BY category
ORDER BY total DESC;
```

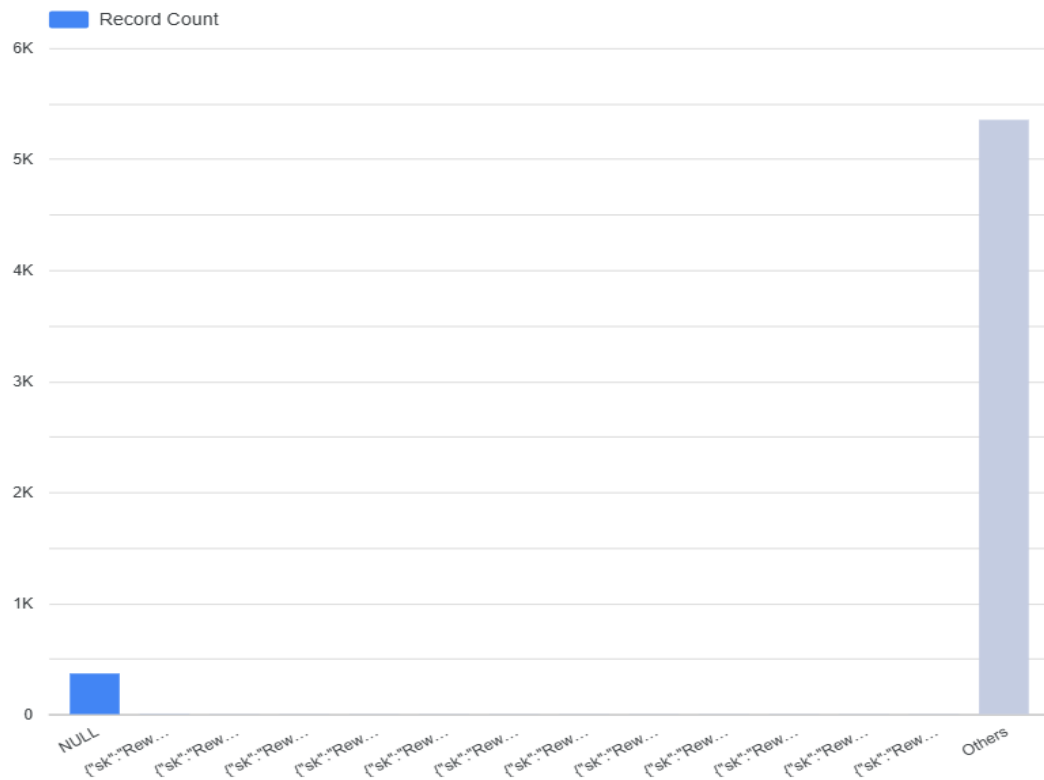


Record Category visualization

Internships make up the largest share of the opportunity record with (1591), followed by Career opportunities (1088), Competitions (734), Courses (632), and Events (622).

4.3 Reward Distribution

```
SELECT "Reward", COUNT (*) AS total
FROM opportunity_data
GROUP BY "Reward"
ORDER BY total DESC;
```

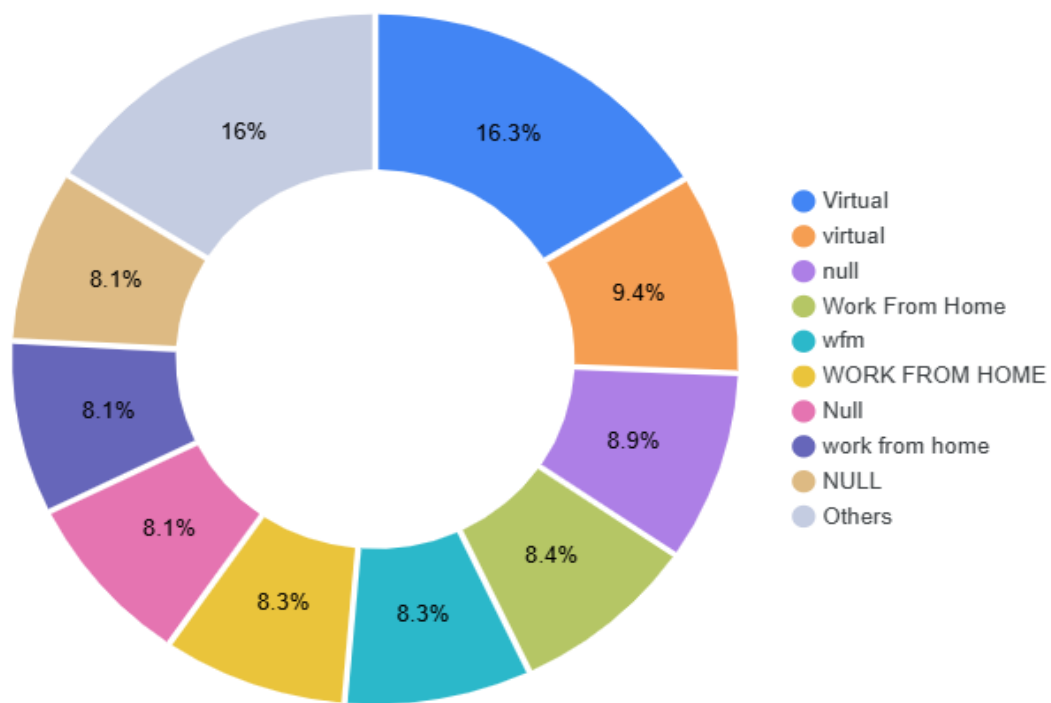


Reward distribution Visualization

4.4 Location distribution

```
SELECT LOWER(TRIM(location)) AS location_clean,
       COUNT(*) AS opportunity_count
FROM opportunity_data
GROUP BY location_clean
ORDER BY opportunity_count DESC
LIMIT 10;
```

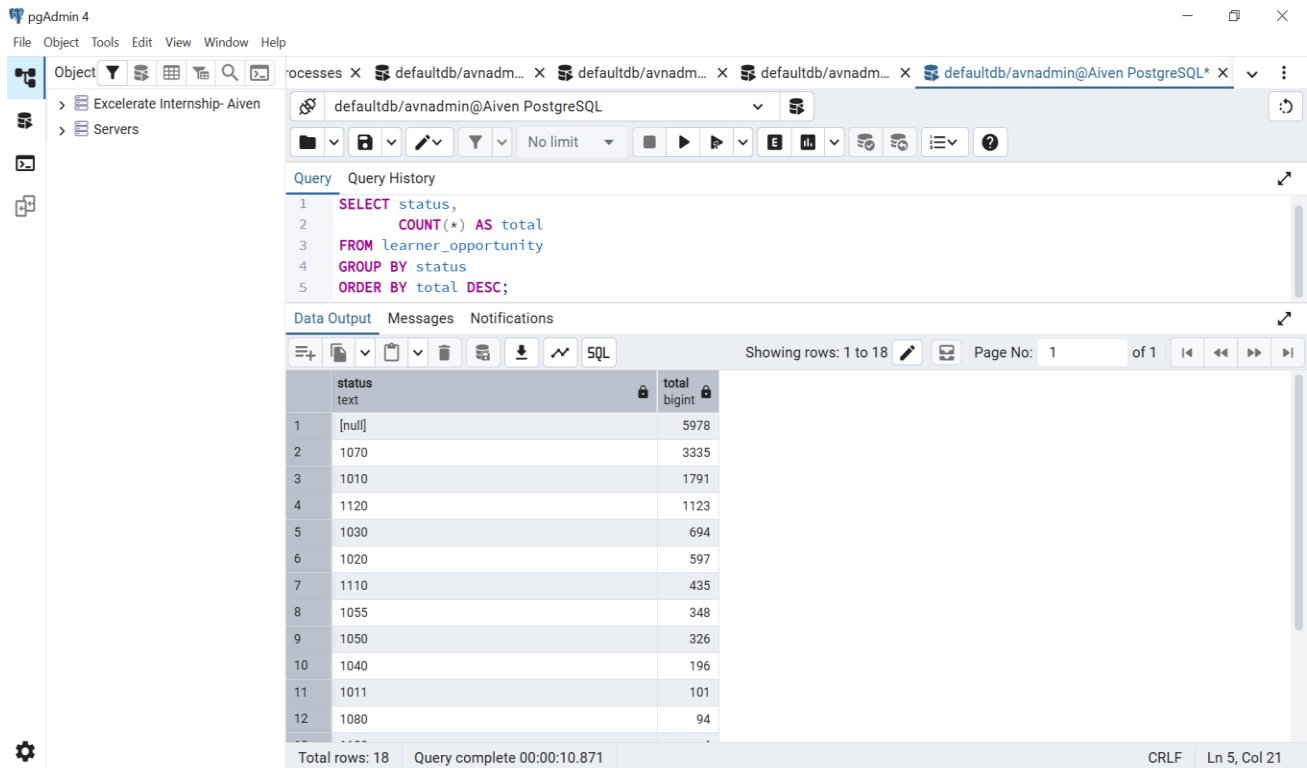
Opportunities are distributed across different locations or delivery modes



Location distribution visualization

4.5 Application Status record

```
SELECT status, COUNT(*) AS total
FROM learner_opportunity
GROUP BY status
ORDER BY total DESC;
```



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the 'Object' tree with 'Excelsior Internship- Aiven' and 'Servers'. The main window shows a SQL query in the 'Query' tab, which is the same query as shown in the code block above. The 'Data Output' tab shows the results of the query, displaying 18 rows. The status values are: [null], 1070, 1010, 1120, 1030, 1020, 1110, 1055, 1050, 1040, 1011, and 1080. The corresponding total counts are: 5978, 3335, 1791, 1123, 694, 597, 435, 348, 326, 196, 101, and 94. The status column is of type 'text' and the total column is of type 'bigint'.

	status text	total bigint
1	[null]	5978
2	1070	3335
3	1010	1791
4	1120	1123
5	1030	694
6	1020	597
7	1110	435
8	1055	348
9	1050	326
10	1040	196
11	1011	101
12	1080	94

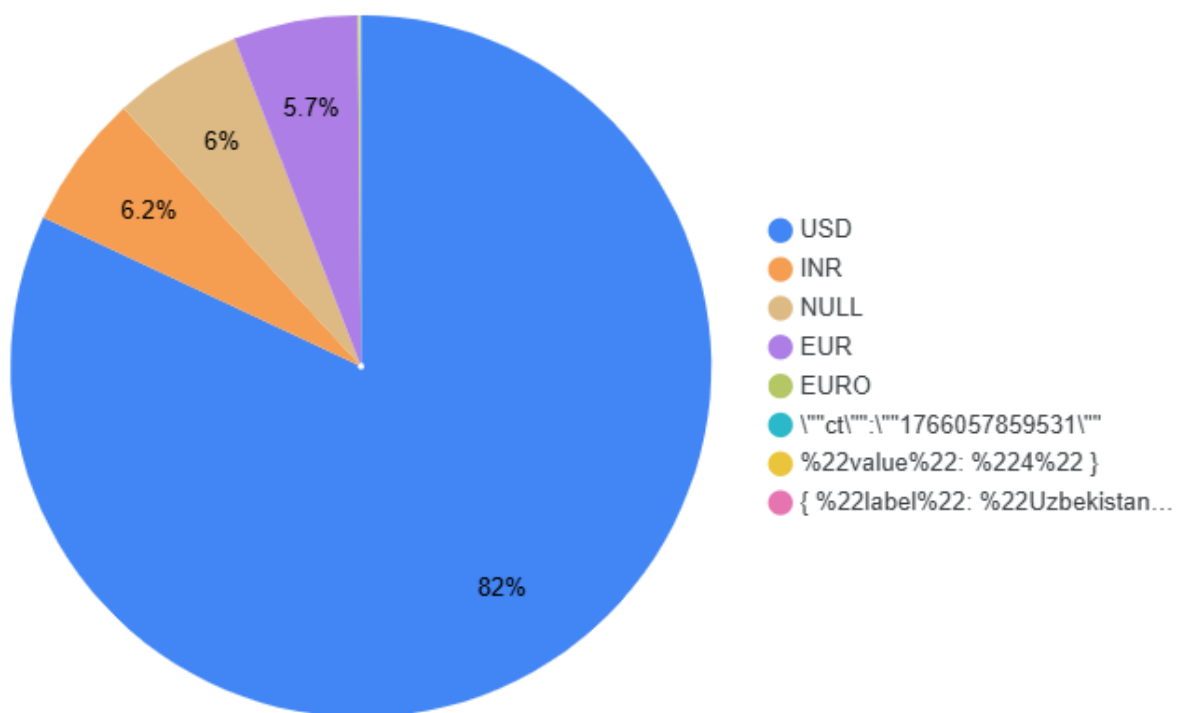
Total rows: 18 Query complete 00:00:10.871 CRLF Ln 5, Col 21

Status record query result

4.6 Currency Record

```
SELECT currency_type,  
       COUNT(*) AS total  
FROM opportunity_data  
GROUP BY currency_type  
ORDER BY total DESC;
```

Multiple currencies exist in the dataset and USD dominates the fee currency field (4,699 records, roughly 82% of populated rows), with INR (355) and EUR (329, plus 5 more recorded as the non-standard “EURO”) making up most of the remainder.



Currency type visualization

4.7 Missing values

```
SELECT
  COUNT (*) FILTER (WHERE status IS NULL) AS missing_status,
  COUNT (*) FILTER (WHERE category IS NULL) AS missing_category,
  COUNT (*) FILTER (WHERE apply_date IS NULL) AS missing_apply_dt,
  COUNT (*) FILTER (WHERE completion_date IS NULL) AS missing_complete,
  COUNT (*) FILTER (WHERE payment_status IS NULL) AS missing_payment
FROM learner_opportunity;
```

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer: Excelsior Internship- Aiven, Servers

Query Editor: defaultdb/avnadmin@Aiven PostgreSQL

```
1 SELECT
2   COUNT(*) FILTER (WHERE status IS NULL) AS missing_status,
3   COUNT(*) FILTER (WHERE category IS NULL) AS missing_category,
4   COUNT(*) FILTER (WHERE apply_date IS NULL) AS missing_apply_dt,
5   COUNT(*) FILTER (WHERE completion_date IS NULL) AS missing_complete,
6   COUNT(*) FILTER (WHERE payment_status IS NULL) AS missing_payment
7 FROM learner_opportunity;
8
```

Data Output: Messages Notifications

Showing rows: 1 to 1 of 1

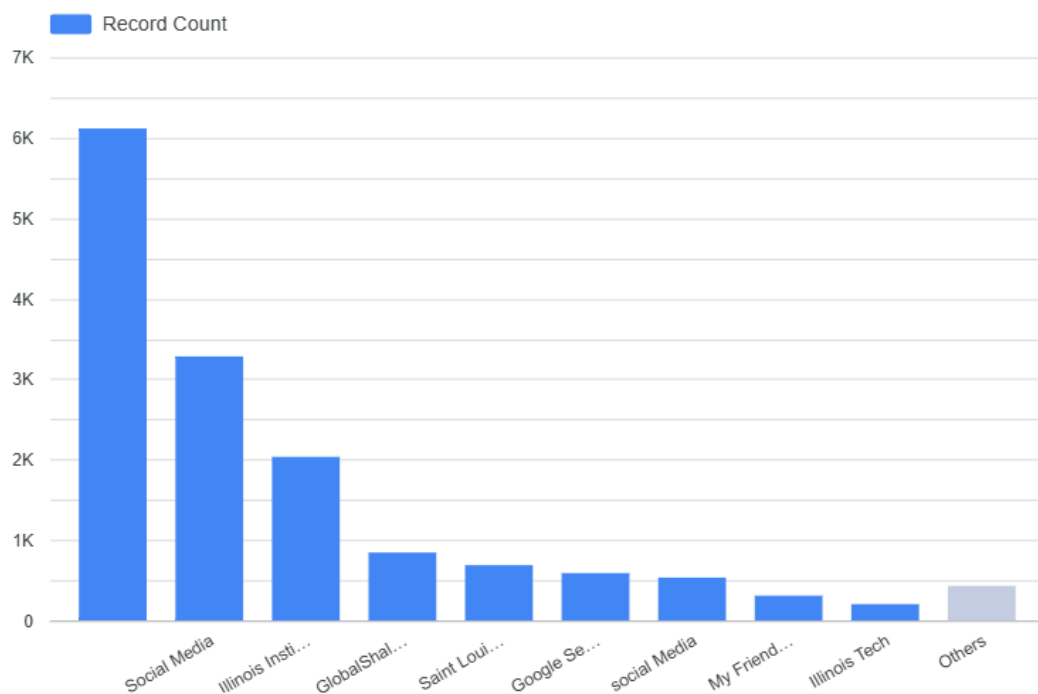
	missing_status bigint	missing_category bigint	missing_apply_dt bigint	missing_complete bigint	missing_payment bigint
1	5978	9279	5982	13897	5995

Total rows: 1 Query complete 00:00:11.822 CRLF Ln 7, Col 25

4.8 Referral Sources

```
SELECT from_where_did_you_hear_about_us,
       COUNT (*) AS total
FROM learner_opportunity
GROUP BY from_where_did_you_hear_about_us
ORDER BY total DESC;
```

Social media is the dominant learners acquisition channel



Referral source visualization

4.9 Duplicate Records

```
SELECT opportunity_id, COUNT(*) FROM opportunity_data
GROUP BY opportunity_id
HAVING COUNT(*)>1;
```

```
SELECT application_id, COUNT(*)
FROM learner_opportunity
GROUP BY application_id
HAVING COUNT(*)>1;
```

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer: Excelerate Internship- Aiven > Servers

Query: defaultdb/avnadmin@Aiven PostgreSQL

Query History

```
1 SELECT application_id,
2 COUNT(*)
3 FROM learner_opportunity
4 GROUP BY application_id
5 HAVING COUNT(*)>1;
```

Data Output Messages Notifications

Showing rows: 1 to 92 Page No: 1 of 1

	application_id text	count bigint
1	[null]	13372
2	12334	2
3	yes yes	2
4	31231313	12
5	64654	20
6	12331	5
7	DASDDA	6
8	365432	2
9	453655	11
10	adfgasdg	2
11	A123cs	7
12	Why do you want to apply for this Internship? *	93

Total rows: 92 Query complete 00:00:01.553 CRLF Ln 5, Col 19

Duplicate query result

5. Exploratory Data Analysis

The initial exploratory analysis was conducted on both the Opportunity Data and Learner Data datasets to understand their structure, content, and overall quality. The analysis revealed several patterns and provided valuable insights into the datasets before any cleaning or transformation.

5.1 Interesting Patterns

- The Opportunity dataset contains a diverse range of opportunity categories, with Internship being the most common category, followed by Career, Competition, Course, and Event.
- The Learner dataset captures information related to learner applications, sign-ups, scholarship status, lead sources, and completion activities, providing a comprehensive view of learner engagement.
- Several fields contain repeated categorical values that are suitable for aggregation and dashboard visualizations.

5.2 Early Trends

- Internship and career-related opportunities account for a significant proportion of the available opportunities.
- Educational and professional development opportunities, including courses, competitions, and masterclasses, also represent a substantial portion of the dataset.
- The learner data indicates that application and engagement information is consistently recorded, making it suitable for future trend analysis.

5.3 Notable Distributions

- Opportunity records are not evenly distributed across categories, with a few categories accounting for the majority of records.
- Some fields contain only a small number of records, indicating less frequently used opportunity types.
- The datasets contain a mix of categorical, numerical, date/time, and text fields, providing flexibility for different types of analysis and visualizations.

5.4 Potential Opportunities for Deeper Analysis

The exploratory analysis identified several areas for future investigation, including:

- Opportunity trends over time.
- Learner application and completion rates.
- Scholarship participation and outcomes.
- Relationships between opportunity characteristics and learner engagement.
- Performance comparisons across different opportunity categories.

6. Data Quality Assessment

During the exploratory analysis, several data quality issues were identified that could affect the reliability of future analysis and reporting.

6.1 Missing Values

Some fields contained NULL or missing values. For example, the category field included records with NULL values, indicating that some opportunities were not assigned a category.

Impact: Missing values can reduce the accuracy of summaries and visualizations. Records with missing categories may be excluded from grouped analyses or produce misleading results.

6.2 Duplicate Records

No obvious duplicate records were identified in opportunity data during the initial exploration. However, a dedicated duplicate check should be performed using a unique identifier such as Opportunity ID before conducting advanced analysis. A full row duplicate check on learner_data identified 171 exact duplicate records with rows identical across every populated field.

Impact:

- Duplicate records can inflate counts and produce inaccurate analytical results.

6.3 Formatting and Capitalization Inconsistencies

The **category** column contained unexpected text values such as encoded JSON strings and inconsistent naming conventions instead of valid category names. Examples observed include encoded text strings, mixed cases, unexpected category labels such as "Xploreu", mixed date formats, and empty columns.

Impact:

- Similar records may be treated as different categories.
- Category counts become inaccurate.
- Dashboards and reports may display incorrect information.

6.4 Mixed Data Types and Unexpected Values

Some categories had extremely low frequencies (e.g., values appearing only once or twice), while others contained invalid or corrupted entries. Examples include :

- Program (2 records)
- NULL (2 records)
- Encoded text value (1 record)

Impact:

- These values may indicate data entry errors or import issues.
- They can distort statistical summaries and reduce confidence in the dataset.

7. Conclusion

Week 1 established a working PostgreSQL environment on Aiven, successfully imported both the Opportunity Data (5,733 records) and Learner Data (15,029 records) datasets with verified, complete row counts, and connected the resulting database to Looker Studio for downstream dashboard work. Initial SQL-based exploration covered dataset size, missing values, duplicates, category and status distributions, referral sources, currency and location breakdowns, and time-based application and completion trends.

That exploration surfaced a specific, actionable set of data quality issues with 171 duplicate records, one structurally corrupted record, inconsistent capitalization across several categorical fields, two incompatible date storage formats.

Taken together, the database is correctly configured and connected, both datasets are understood at a structural level, and a clear, prioritized cleaning and validation plan is in place for Week 2. This report, together with the SQL queries in Section 4, provides the working reference needed to move directly into normalization, relationship building, and the first dashboard iteration.